

ECE250: Signals & Systems Quiz-3

Date: 10-September-2025

Total Points: 10

Duration: 15 Minutes

Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
 2. Institute Plagiarism policy are strictly applicable.
-

[CO1] [Q1]: Let

$$x(t) = u(t - 3) - u(t - 5) \text{ and } h(t) = e^{-3t}u(t)$$

- (a) Compute $y(t) = x(t) * h(t)$. **[4 Points]**
- (b) Compute $g(t) = (\frac{dx(t)}{dt}) * h(t)$. **[4 Points]**
- (c) How is $g(t)$ related to $y(t)$? **[2 Points]**